SMARTLEARN AI - ACADEMIC SYSTEM

Program: BSc Computer Science | *Course Materials*

Course Description & Objectives

This course introduces the fundamental concepts of computer programming and software development. Students will learn how to design, write, test, and debug programs using high-level programming languages. Key focus areas include procedural programming, control structures, basic data structures, and the fundamentals of object-oriented programming.

By the end of this course, students will be able to:
- Solve computational problems by writing clear, correct, and well-structured code.
- Understand variables, data types, logic control, loops, and lists.
- Implement modular functions and reuse code.
- Read and write to external files.

Module 1: Variables, Operators, and Expressions

A variable is a named storage location in memory. It has a data type which determines the size and layout of the variable's memory. Common data types include Integers, Floats, Strings, and Booleans. Operators allow operations on variables (arithmetic, relational, logical).

Example Python snippet:

```
x = 10
y = 5.5
sum_result = x + y
print('The sum is:', sum_result) # Output: 15.5
```

Module 2: Flow Control (Conditionals & Loops)

Control statements dictate the execution order of individual statements in a program. If-else statements introduce branching based on conditional checks. For and while loops introduce iteration to run a code block multiple times.

Key constructs include:
- If, Elif, Else checks.
- While loop (runs while condition is true).
- For loop (iterates over sequences like lists or ranges).